EE6094
CAD for VLSI Design

Workstation User's Guide

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Slides Credit: TA Shih-Cheng Huang / TA Yi-Ting Lin / TA Yin-Rong Zhuo

Outline

- VPN
- Workstation
- Vim
- Tmux

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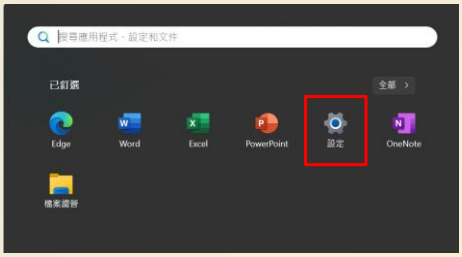
- VPN
- Workstation
- Vim
- Tmux

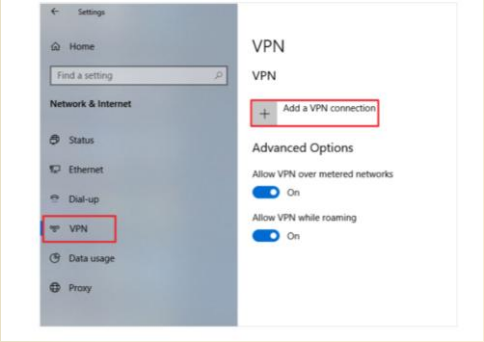
VPN

- To prevent the leakage of advanced semiconductor manufacturing technology, we restrict access via VPN
  - It's your responsibility to protect advanced semiconductor manufacturing technology data
- How to connect to VPN
  - Go to settings



VPN

- How to connect to VPN
  - Settings > Network & Internet > VPN
  - Add a VPN connection



VPN

- How to connect to VPN
  - VPN提供者: Windows(內建)
  - 連線名稱:自訂
  - 伺服器: 140.115.71.148
  - VPN類型:
    L2TP/Ipsec(使用預先共用金鑰)
  - 共用金鑰: 123
  - 使用者名稱: 見下頁分配
  - 密碼:見下頁分配

VPN

- How to connect to VPN

ID range	使用者名稱	密碼
109401003 - 110504515	CADeeNCU1	AndyChen0401
113226008 - 113521152	CADeeNCU2	AndyChen0402
113522041 - 113827030	CADeeNCU3	AndyChen0403

Outline

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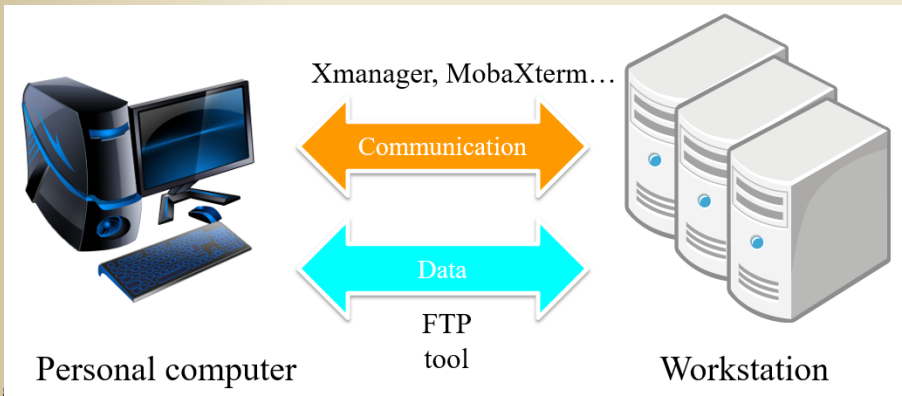
Workstation

- Introduction of workstation

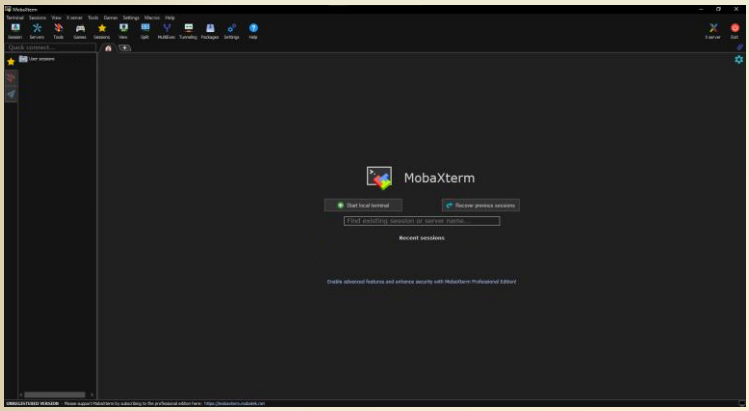


Workstation

- You are requested to complete most of your programming assignments on our server
  - And we will evaluate your program with the server
- Server setting
  - System: Linux 3.10.0
  - Host: 140.115.71.87
  - Port: 22 (Please use SSH to connect to the machine)
    - Default account: s(Your Student_ID)
      - Ex. s123456789
    - Default password: ncuedaCAD
    - The first login requires a mandatory password change

Workstation

- How to connect to a server with SSH Client?
  - MobaXterm

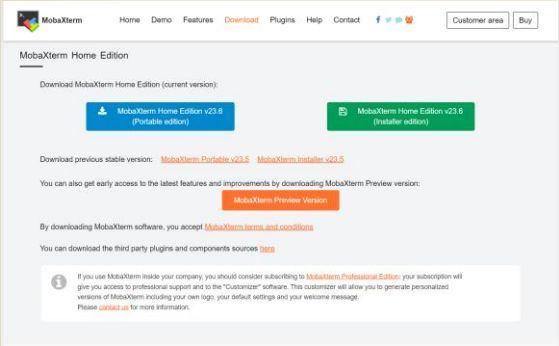


Workstation

- MobaXterm
  - MobaXterm is your ultimate toolbox for remote computing.
  - MobaXterm provides all the important remote network tools (SSH, X11, RDP, VNC, FTP, MOSH, ...) and Unix commands (bash, ls, cat, sed, grep, awk, rsync, ...) to Windows desktop
  - There are many advantages of having an All-In-One network application for your remote tasks

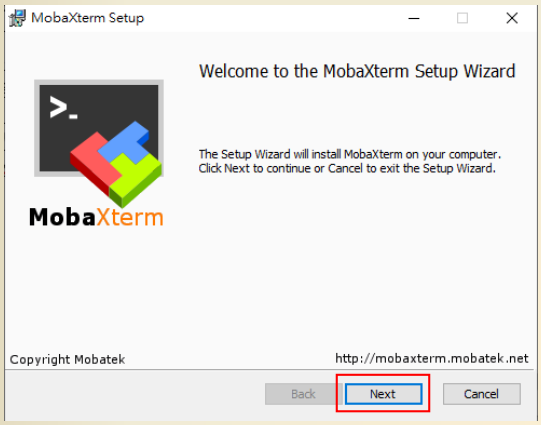
Workstation

- How to connect the Linux server
  - Step 1: download MobaXterm https://mobaxterm.mobatek.net/download-home-edition.html



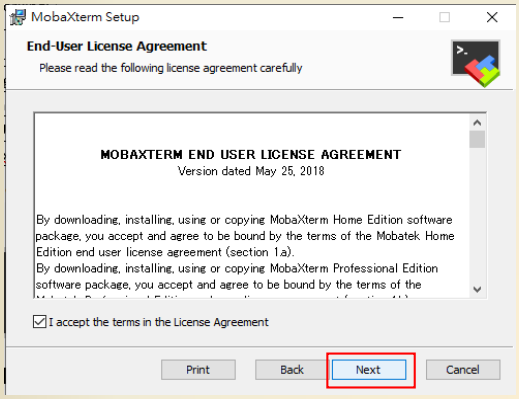
Workstation

- How to connect Linux server
  - Step 2-1: install MobaXterm



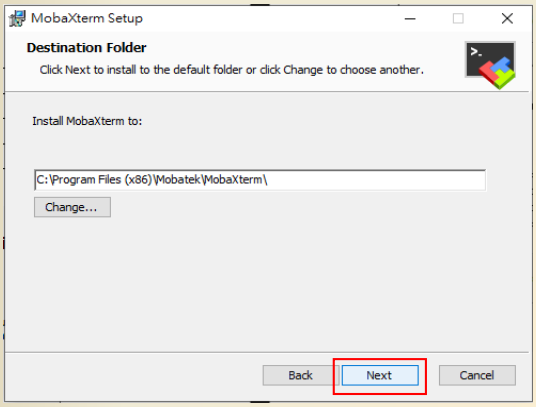
Workstation

- How to connect Linux server
  - Step 2-2: install MobaXterm



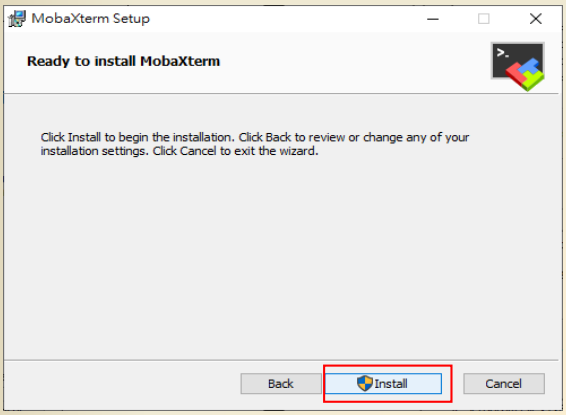
Workstation

- How to connect Linux server
  - Step 2-3: install MobaXterm



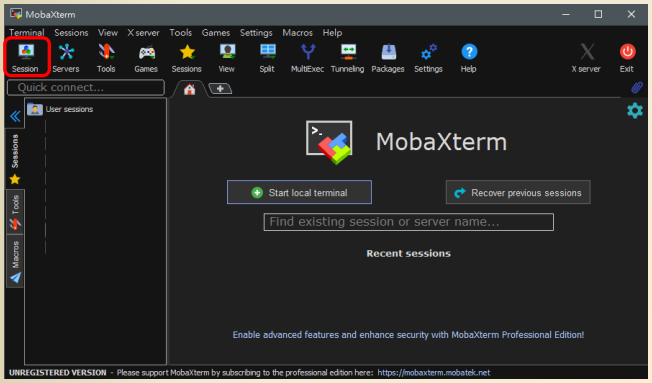
Workstation

- How to connect Linux server
  - Step 2-4: install MobaXterm



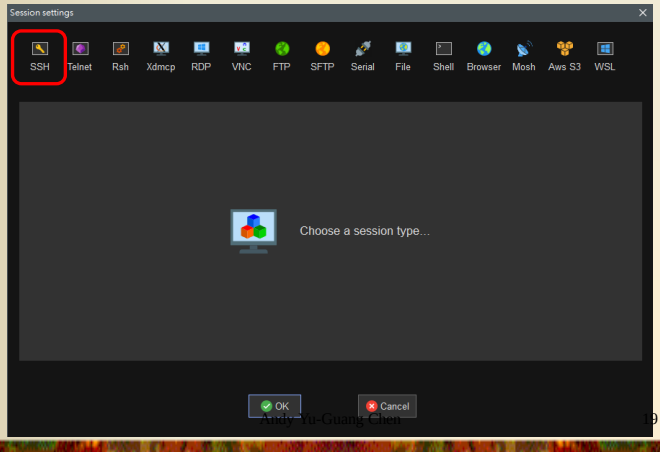
Workstation

- How to connect Linux server
  - Step 3-1: Click session to create a new session



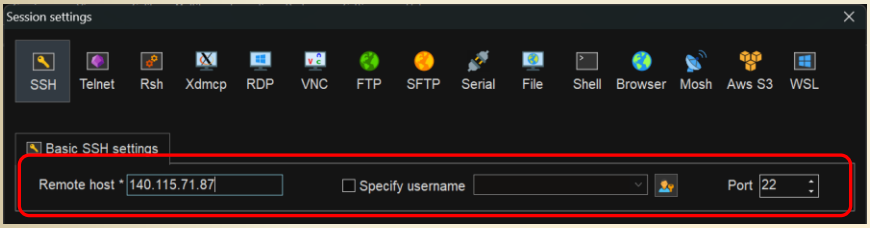
Workstation

- How to connect Linux server
  - Step 3-2: Click SSH to create a new SSH session



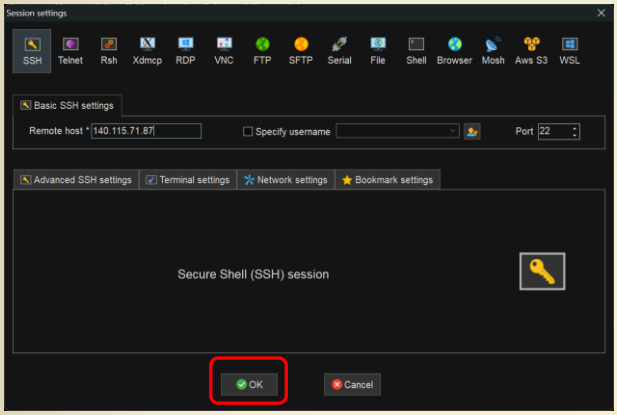
Workstation

- How to connect Linux server
  - Step 3-3: Fill in the given Host to the Host field and set the port number to 22



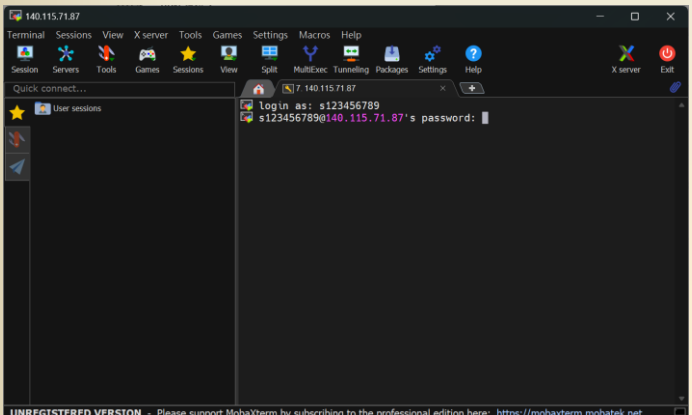
Workstation

- How to connect Linux server
  - Step 3-4: Click ok



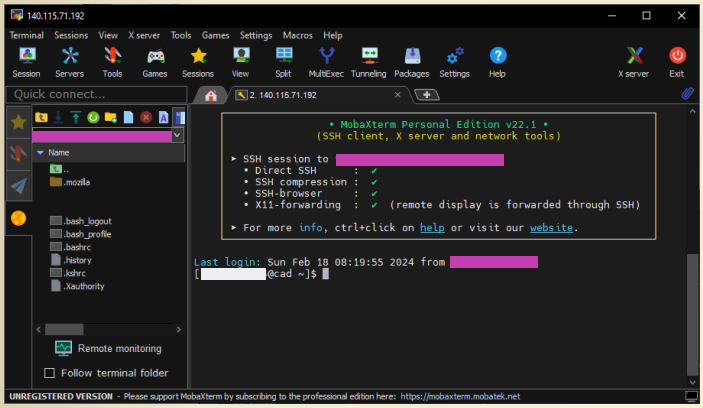
Workstation

- How to connect Linux server
  - Step 3-5: Login



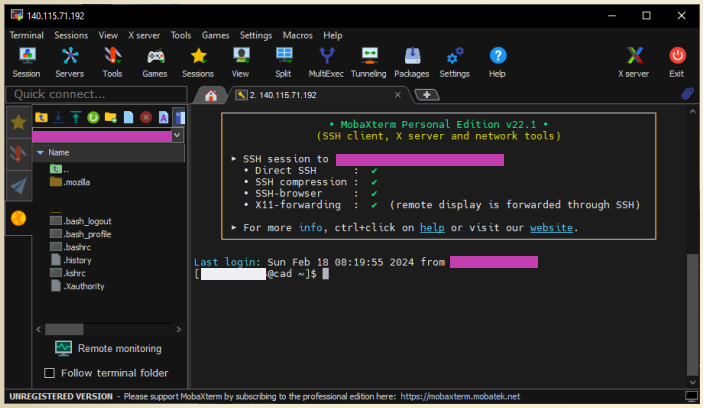
Workstation

- How to connect Linux server
  - Step 3-6: Login



Workstation

- How to connect Linux server
  - Step 3-6: Login



Workstation

- How to connect Linux server
  - Step 3-7: The first time login, you must change password

```
You are required to change your password immediately (root enforced)
[redacted]
WARNING: Your password has expired.
You must change your password now and login again!
Changing password for user [redacted]
Changing password for [redacted]
(current) UNIX password:  → Default password
New password:             → New password
Retype new password:      → New password
[redacted]
passwd: all authentication tokens updated successfully.
```


Basic command of workstation

- ls (list)
- ll (long list format)
- cd (change directory)
- pwd (print working directory)
- cp (copy)
- mv (move)
- rm (remove)
- mkdir (make directory)
- rmdir (remove directory)
- tar (compression tool)
- passwd (password)
- Ctrl + c (force quit)
- ps (process status)
- kill (kill process)

Basic Command

- mkdir

```
[[redacted]@cad ~]$ mkdir PA1
[[redacted]@cad ~]$
```

- ls

```
[[redacted]@cad ~]$ ls
[redacted] PA1  PA2  PA3  PA4
```

- cd

```
[[redacted]@cad ~]$ cd PA1
[[redacted]@cad ~/PA1]$
```


Basic Command

- cp

```
[[redacted]@cad ~/PA1]$ cp ../test.cpp PA1.cpp
```

- rm

```
[[redacted]@cad ~/PA1]$ ls
PA1.cpp
[[redacted]@cad ~/PA1]$ rm PA1.cpp
[[redacted]@cad ~/PA1]$ ls
[[redacted]@cad ~/PA1]$
```

Basic Command

◆rmdir

```
[[redacted]@cad ~]$ ls
[redacted] PA1  PA2  PA3  PA4  test.cpp
[[redacted]@cad ~]$ rmdir PA1
[[redacted]@cad ~]$ ls
[redacted] PA2  PA3  PA4  test.cpp
[[redacted]@cad ~]$
```


Basic Command

◆mv

```
[[redacted]@cad ~/PA1]$ ls
README.md [redacted]
[[redacted]@cad ~/PA1]$ ls ../
[redacted] PA1  PA2  PA3  PA4  test.cpp
[[redacted]@cad ~/PA1]$ mv ../test.cpp ./
[[redacted]@cad ~/PA1]$ ls
README.md  test.cpp
[[redacted]@cad ~/PA1]$ ls ../
[redacted] PA1  PA2  PA3  PA4  [redacted]
[[redacted]@cad ~/PA1]$
```

Workstation

- Change your workstation password
  - Step 1: Key in “ passwd ”
  - Step 2: Key in your current password
  - Step 3: Key in your new password
  - Step 4: Retype your new password again
  - The info with “**successfully**” means password change is success

```
[[redacted]@cad ~]$ passwd  Step1
Changing password for user [redacted].
Changing password for [redacted].
(current) UNIX password: Step2
New password: Step3
Retype new password: Step4
passwd: all authentication tokens updated successfully.
```


Outline

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- Tmux

Vim

- It is an efficient text editor especially developed for Linux users, that it is mainly used to edit or create different types of files
- Vim is the most popular and extremely powerful text editor
  - It possesses a lot of features that you would not expect to have in a text editor

```
[[redacted]@cad ~/PA1]$ vim parser.cpp
```

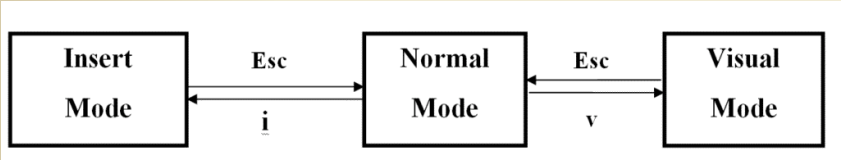
Open parser.cpp or create a new file named parser.cpp

Vim

- The Vim editor is a modal text editor
  - It uses modes for different purposes like inserting text, running commands, and selecting text
- There are three modes of operation
  - Normal
    - The initial mode of the Vim editor
    - Normal mode is also known as command mode because all the keystrokes you perform are interpreted as commands
  - Insert
    - Insert mode is where you can insert your text in the file
    - This mode inserts every character you type at the current cursor location
  - Visual
    - Visual mode allows you to select text so that you may perform certain operations (cut, copy, delete) on it

Vim

- Changing the modes



Vim

- Normal mode
  - You will see the below screen after executing the command
  - This is your normal mode in Vim

```
~
~
~
"parser.cpp" [New File]
```

Vim

- Insert mode
  - You should be in the Insert mode if you want to **edit** your file
  - Press “i”, “a” or “o” from your keyboard, and you will be in insert mode
  - Press Esc to back to normal mode

```
#include <string>

void main() {
}
~
-- INSERT --
```


Vim

- Saving your work
  - When you are in normal mode, press “:w” to save your work and press “:q” to exit vim
  - You also can use “:wq” to save and exit vim

```
#include <string>

void main() {
}
~
:wq
```

Vim

◆File related commands

:w	write the file to the disk
:q	quit vi without saving the file
:wq	write the file to disk and quit vi
:q!	ignore the warning and discard the change
:w filename	save the file as filename

Vim

◆Moving the cursor

j	move the cursor down one line
k	move the cursor position up one line
l	move the cursor to the bottom of the screen
0	move to the beginning of the line
$	move to the end of the line

Vim

◆Inserting Text

I	insert text at the beginning of the line
i	insert text before the current cursor location
a	insert text after the current cursor location
o	create a new line for the text below the current cursor location
O	create a new line for text above the current cursor location

Vim

◆Changing text

cc	remove the whole line and start Insert mode
s	remove the character under the cursor and start Insert mode
r	replace the character under the cursor

Vim

◆Copying pasting

y	copy the selected text to clipboard
yy	copy current line
P	insert the text “before” the cursor
p	insert the text at the point after the cursor

Vim

◆Deleting Text

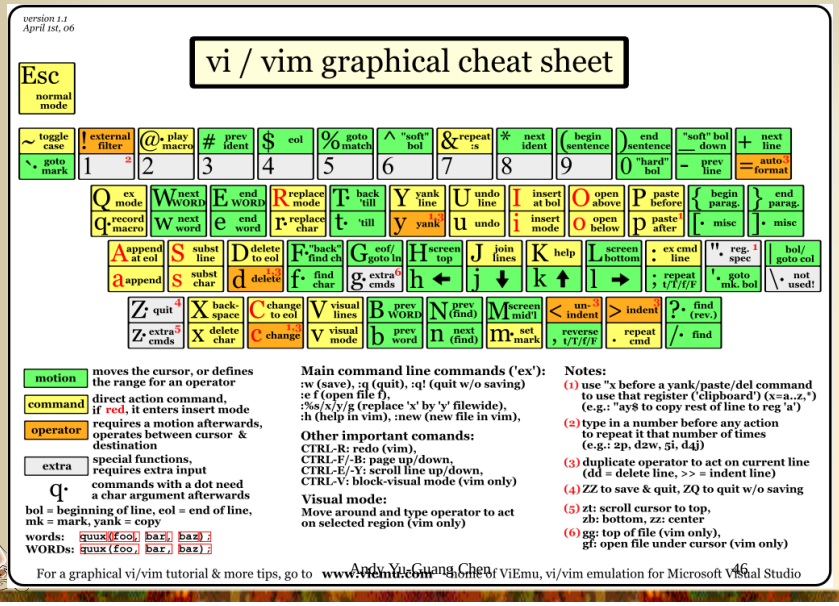
X	delete the character before the current location
x	delete the character under the current location
D	cut to the end of line
dd	cut current line

Vim

◆Undo/Redo

u	undo last change
Ctrl+R	redo

Vim



Vim

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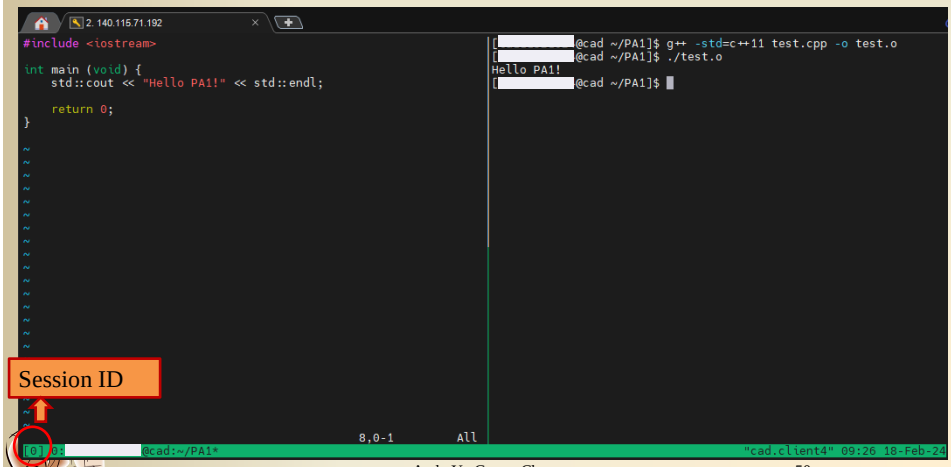
Tmux

- What is tmux?
  - Tmux is a terminal multiplexer that you can start a Tmux session and then open multiple windows inside that session
  - Session ➔ windows

```
[[redacted]@cad ~/PA1]$ tmux
```


Tmux

- A tmux session with two windows



Tmux

- Working with tmux windows and panes
  - Ctrl+b c: Create a new window.
  - Ctrl+b w: Choose window from a list.
  - Ctrl+b 0: Switch to window 0.
  - Ctrl+b n: Switch to next window.
  - Ctrl+b p: Switch to previous window.
  - Ctrl+b %: Split current pane horizontally into two panes.
  - Ctrl+b “: Split current pane vertically into two panes.
  - Ctrl+b arrow keys: Switch pane.

Reference

- Linux
  - https://files.fosswire.com/2007/08/fwunixref.pdf
  - https://linux.vbird.org/
- Vim
  - https://danielmiessler.com/study/vim/
  - http://www.vixual.net/blog/archives/234
- Tmux
  - https://blog.gtwang.org/linux/linux-tmux-terminal-multiplexer-tutorial/
  - https://linuxize.com/post/getting-started-with-tmux/#starting-your-first-tmux-session